

Radioactivity Studies in Antarctica

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Abstract

The variation of ^7Be , ^{210}Pb , ^{137}Cs and total beta activity has been studied in the Antarctic shallow ice core samples upto 9m depth with reference to their activities in fresh snow, surface ice samples and fresh lakewater samples collected near the Indian station, Dakshin Gangotri (70°S , 12°E). Based on the ^{210}Pb , ^{137}Cs and total beta activity measurements in shallow ice core samples, past accumulation rate of snow has been estimated to be $13.5\text{g}/\text{cm}^2\text{yr}$ ($0.3\text{m}/\text{yr}$).

Introduction

Antarctica the coldest desert of the earth, receives very little annual precipitation of 5 cm in the form of snow which gradually increases upto 50 cm near coastal areas. Direct measurements of precipitation and accumulation rates of ice in Antarctica are very difficult and often complicated due to strong blizzards which throw large amounts of loose snow into the atmosphere, upto 100 metres, causing a whiteout and gets redeposited in some regions depending upon the wind directions. Generally, the precipitation and blizzard occur simultaneously and it is difficult to measure the precipitation component separately. Indirect methods, therefore, based on measurements of seasonal variations of $\delta^{18}\text{O}$ and radioactive tracers of suitable half-lives are employed for dating the annual layers of ice and estimation of past accumulation rates of ice in Antarctica.

Large number of natural radioisotopes (like ^3H , ^{32}Si , ^{14}C , ^{210}Pb) and artificial radioisotopes (like ^{60}Co , ^{90}Sr , ^{137}Cs , ^{239}Pu) are produced in the atmosphere from i) the interaction of cosmic rays with the gases from the atmosphere like nitrogen, argon; ii) the uranium decay series gaseous products like radon gas by degassing of the earth's crust which further decays into ^{210}Pb , ^{210}Bi and ^{210}Po ; and iii) the testing of nuclear weapons during past 4 decades.

These isotopes, depending on their half-lives, have been used as tracers for studying different geophysical processes of different time spans.

For the present work, three radioisotopes ^7Be ($t_{1/2}=53\text{d}$, γ emitter), ^{210}Pb ($t_{1/2} = 22.3\text{ yr}$, beta emitter) and ^{137}Cs ($t_{1/2} = 30\text{ yr}$, γ emitter) were chosen, since their half-lives are suitable for dating of glacier ice upto 100 years and estimation of accumulation rate of ice near the Indian station, Dakshin Gangotri (70°S , 12°E) in Antarctica. The total beta activity produced both by natural and artificial sources which can be used as time-marker in identification of nuclear horizon produced during the past 4 decades has also been measured.

Materials and Methods

With a view to observe and compare the present day radioactivity levels in fresh snow with that of ice deposited during past 3 decades, one snow, two fresh lakewater samples and several surface as well as shallow ice samples from the ice face near the Indian Bay, shallow ice core at Dakshin Gangotri (DG) Station, a floating iceberg near the Indian Bay and one old ice sample from the glacier located west of Maitree hills were collected. Of the total 54 samples, total beta activity was measured in 52 samples, ^{210}Pb activity in 32 samples, ^7Be in eight samples (Tables I, II, III and IV) whereas ^{137}Cs activity was measured only in selected 14 samples (Table V). Volumes of the samples collected from ice cores and surface ice samples ranged from 0.2 to 1.7 L whereas for the other samples, it ranged from 4 to 25 L.

The samples were collected in leak-proof plastic bottles and transported to the laboratory for analysis. A few samples were processed at the sampling site wherever necessary (in case of volumes $>4\text{ L}$) and the $\text{Fe}(\text{OH})_3$ matrix which scavenged the radioactivity from the samples (as discussed by Nijampurkar *et al.*, 1984) were packed and brought to the laboratory for radiochemical analysis.

Chemical procedures for radiochemical separation and purification of ^{210}Pb and non-destructive analysis of ^{137}Cs , total beta and ^7Be by the assay of beta and gamma activities are discussed in detail elsewhere (Nijampurkar *et al.*, 1982; Bhandari *et al.*, 1984).

Results and Discussion

The measurements of natural and artificial radioactivity, made in terms of ^7Be , ^{210}Pb , ^{137}Cs and total beta activity on fresh snow, surface ice and lakewater samples in Antarctica give the present day radioactivity levels in precipitation at DG Station whereas the shallow ice samples collected at different depths give the variation of snow deposition during last 3-4 decades. Artificial radioactivity (^{137}Cs and total beta activity) measurements give clue in estimation of past accumulation rate of ice near the Indian Station in Antarctica.

- i) ^7Be : The half-life of ^7Be being 53 days, it gives useful information about

the samples which are as old as 6-8 months knowing the mean values of ^7Be in fresh precipitation at that location. Table I shows the ^7Be activity in fresh snow precipitation (Sample B) to be 8.8 dpm/L whereas all other samples of surface ice and lakewaters show ^7Be activities to be generally <1 dpm/L which suggest that these samples are older than 6 months. As expected, the sample 'Q' collected from snout ice of Maitree glacier which is supposed to be very old, does not show any ^7Be activity.

ii) ^{210}Pb : Lead-210 is one of the most useful tracers for dating ice upto 100 years and has been often used for estimation of past deposition rates of ice in polar as well as tropical glaciers (Croaz and Langway, 1966; Gaggler *et al.*, 1983; Nijampurkar *et al.*, 1982).

The ^{210}Pb values of 17 snow and surface ice samples (Tables I, II, III and IV) vary from 10 to 130 dph/L with a mean value of 50 dph/L which is lower by a factor of 3 to 5 than that observed in Greenland, Alps and Himalayas (Croaz and Langway, 1966; Gaggler *et al.*, 1983; Nijampurkar *et al.*, 1982). The lower values of 10-20 dph/L can be attributed to the seasonal variations and

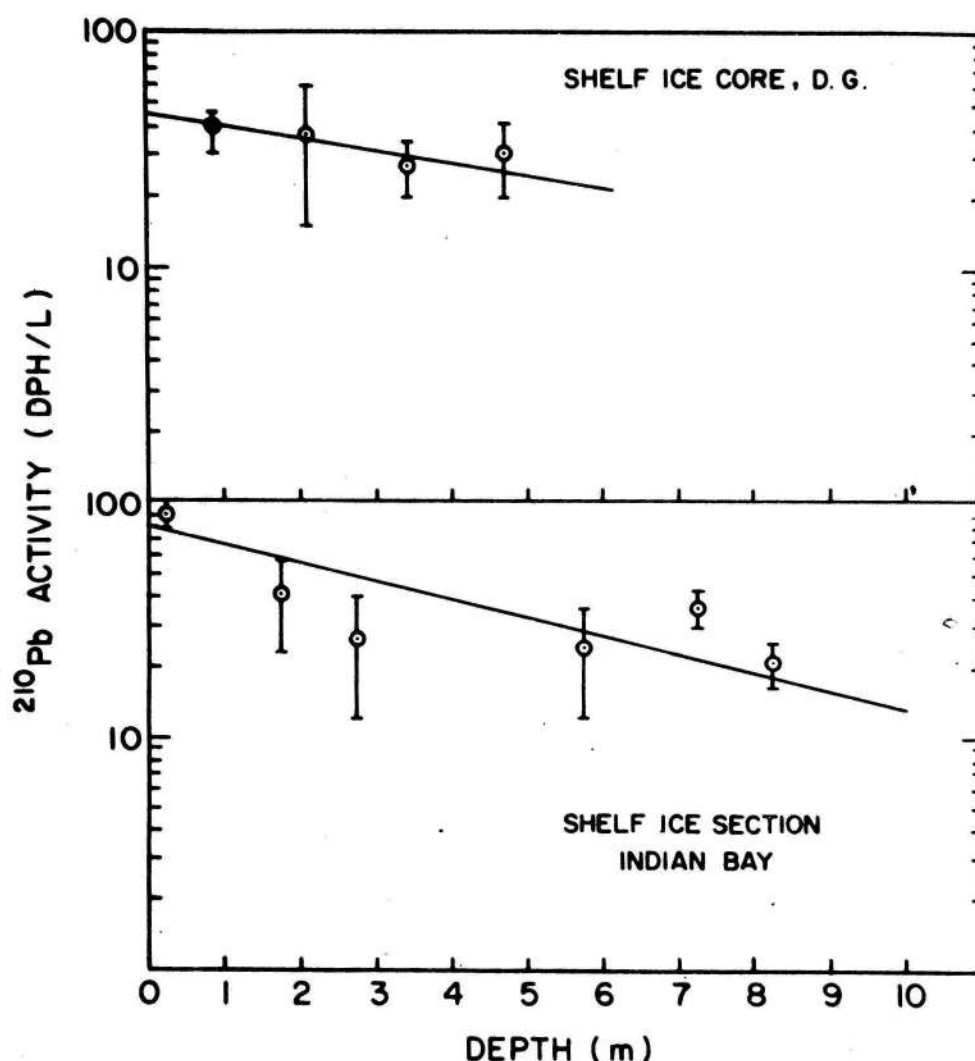


Fig 1. The depth profile of ^{210}Pb activity (dph/L) of the shallow ice core raised from the shelf ice near D.G. Station and Shelf Ice face section near the Indian Bay

Table I. Total beta, ^{210}Pb and ^7Be activities in few miscellaneous samples from Antarctica

Code No.	Nature	Volume (L)	Total beta activity (dph/L)	^{210}Pb activity (dph/L)	^7Be activity (dpm/L)
B	Windborne snow	25	122.36 ± 12.88	37.14 ± 3.72	8.8 ± 0.2
C	Shelf ice (DG)	25	83.67 ± 11.04	17.04 ± 1.56	1.1 ± 0.08
D	Shelf ice (Indian Bay)	15	62.74 ± 9.48	35.16 ± 3.24	0.18 ± 0.02
E	Iceberg	14	318.68 ± 32.08	15.78 ± 3.42	0.97 ± 0.07
P	Maitree Lake Water	20	72.56 ± 9.48	9.54 ± 2.58	0.07 ± 0.02
Q	Snout ice (West Maitree hills)	25	141.59 ± 12.32	9.84 ± 0.96	B.D.L.
R	DG Snout Ice (surface meltwater, East Maitree hills)	13	—	19.80 ± 1.98	0.29 ± 0.03
W	Frozen Lakewater (Wohlthat)		4.50 ± 1.14	0.57 ± 0.07	B.D.L.

BDL = Below detection limit.

Table II. Total beta and ^{210}Pb activities in surface ice samples from Antarctica

Code No	Distance from Shore (km)	Volume (L)	Total beta activity (dph/L)	^{210}Pb activity (dph/L)
Y-0	0.0	1.64	62.74 ± 9.48	35.16 ± 3.24
Y-1	2.0	0.30	140.20 ± 46.97	—
Y-2	4.0	0.32	100.20 ± 30.86	68.93 ± 29.35
Y-3	6.0	0.67	170.70 ± 27.11	—
Y-4	8.0	0.62	91.00 ± 19.97	19.48 ± 13.45
Y-5	10.0	0.71	661.40 ± 82.78	—
Y-6	12.0	0.63	454.15 ± 47.62	39.19 ± 42.15
Y-7	14.0	0.28	383.50 ± 65.37	—
Y-8	16.0	0.85	90.00 ± 7.60	38.48 ± 8.11
Y-9	19.0	0.70	48.00 ± 9.00	—
Y-10	22.0	0.80	249.00 ± 19.20	72.31 ± 9.42
Y-11	25.0	0.51	612.00 ± 42.00	—
Y-12	28.0	0.55	144.00 ± 18.00	107.30 ± 37.29
Y-13	31.0	0.80	112.80 ± 12.60	—
Y-14	34.0	1.00	94.80 ± 10.80	8.97 ± 24.85

large fluctuations observed in single precipitation events as observed by many researchers at different locations on the earth. In spite of these large variations, it is possible to use the mean of the surface activities of that location for calculating the possible accumulation rates of ice at that location. The lower ^{210}Pb values of the lakewaters P, R and W (Table I) can be explained by mixing of young waters with that of old ice melt water contributions of the Maitree and Wohlthat glaciers.

Table III. Total beta and ^{210}Pb activities in Shelf Ice Core from Indian Bay, Antarctica

Code No.	Depth (m)	Volume (L)	Total beta activity (dph/L)	^{210}Pb activity (dph/L)
N-1	Surface	1.64	79.76 ± 7.76	51.53 ± 15.05
N-2	0-0.5	1.12	428.47 ± 34.59	138.37 ± 15.40
N-3	0.5-1.0	0.95	110.80 ± 11.30	—
N-4	1.0-1.5	1.25	134.80 ± 12.70	—
N-5	1.5-2.0	0.75	105.17 ± 10.59	39.93 ± 17.45
N-6	2.0-2.5	0.90	75.50 ± 11.30	—
N-7	2.5-3.0	0.73	95.30 ± 11.11	—
N-8	3.0-3.5	1.38	125.60 ± 16.94	—
N-9	3.5-4.0	1.05	103.76 ± 11.30	26.31 ± 14.77
N-10	4.0-4.5	1.15	116.47 ± 11.99	—
N-11	4.5-5.0	1.05	67.05 ± 10.59	—
N-12	5.0-5.5	1.63	95.30 ± 13.40	—
N-13	5.5-6.0	1.18	174.86 ± 17.94	23.80 ± 11.56
N-14	6.0-6.5	1.60	233.63 ± 30.66	—
N-15	6.5-7.0	0.85	257.40 ± 27.57	—
N-16	7.0-7.5	1.54	108.57 ± 13.69	36.42 ± 7.28
N-17	7.5-8.0	1.45	122.75 ± 13.44	—
N-18	8.0-8.5	1.52	238.80 ± 28.65	20.69 ± 4.51
N-19	8.5-9.0	1.39	163.11 ± 13.08	—

The ^{210}Pb activities of the ice face samples (collected upto 9m depth) from shelf ice near the Indian Bay and shallow ice core at DG ice shelf (upto 5 metres) and iceberg are given in Tables III and IV and plotted in Figs. 1 & 2. The decay of ^{210}Pb activity from two cores as observed in the figure estimates the past accumulation rate of ice to be ~ 0.3 m/yr i.e. $13.5 \text{ g/cm}^2 \text{ yr}$ assuming

Table IV. Total beta and ^{210}Pb activities in iceberg and shelf ice shallow cores of Dakshin Gangotri Station, Antarctica

Code No	Depth (m)	Volume (L)	Total beta activity (dph/L)	^{210}Pb activity (dph/L)
Iceberg:				
F-1	0-0.3	1.00	318.68 ± 32.08	15.3 ± 4.28
F-2	0.3-0.8	0.53	201.60 ± 22.80	53.49 ± 17.75
F-3	0.8-1.5	1.13	98.40 ± 12.60	24.16 ± 8.33
F-4	1.5-1.8	0.68	465.00 ± 31.20	50.59 ± 18.52
Shelf Ice:				
G-1	0-0.6	1.65	83.67 ± 11.04	17.04 ± 1.56
G-2	0.6-1.2	0.75	174.60 ± 36.00	40.48 ± 10.17
G-3	1.2-1.8	1.60	126.00 ± 12.00	—
G-4	1.8-2.4	1.35	45.00 ± 10.20	36.88 ± 22.13
G-5	2.4-3.0	1.61	85.20 ± 9.60	—
G-6	3.0-3.8	1.59	60.60 ± 9.00	27.21 ± 7.08
G-7	3.8-4.4	1.58	52.80 ± 7.20	—
G-8	4.4-5.0	1.36	168.00 ± 10.80	31.33 ± 11.57

an average density of 0.45 gm/cm^3 upto 9m depth covering a period of three decades in this region.

The ^{210}Pb activities in an ice profile of 9m depth sampled at regular interval of 0.5m (Table III) shows three peaks corresponding to samples N-2, N-15 and N-18. The two peaks N-15 and N-18 may be related to 1962-63 and 1958-59 nuclear tests whereas the peak at N-2 (surface sample) is intriguing as discussed in iii) a.

iii) ^{137}Cs and total beta activity:

a) Snow and surface ice samples

^{137}Cs and total beta activity produced in the southern hemisphere due to formation of artificial radionuclides during the testing of nuclear weapons in past four decades is scavenged by the fresh precipitation, gets deposited all over the Antarctic icesheet and is well-preserved layer by layer. These activities are often used as time-markers for calculating the past accumulation rates of ice if the nuclear horizon is identified in the deeper ice samples. It is known from the earlier work that the total beta activity deposited in Antarctica, as a result of Russian nuclear test in 1962 is of the order of 500 dph/kg of ice with a background of $100 \pm 50 \text{ dph/kg}$.

In the fresh snow sample B, collected in January, 1986, the total beta activity level has been observed to be 122 ± 13 dph/kg whereas the samples C, D and P show total beta activity to be <100 dph/kg which is close to the background value suggesting that no atmospheric nuclear tests were conducted during 1983-84.

A set of surface ice samples collected at every 2km interval away from the coast upto 34km show an average of 250 ± 50 dph/kg. High values of 500 and 430 dph/kg are observed at two different locations at a distance of 12-14 km away from the coast near the DG station and 25 km away from the coast. Perhaps the blizzards have dumped large quantities of loose snow during 1983-84 at this location which has high activity. Such high activities observed in the surface samples are intriguing and difficult to explain. However, it may be possible that such activity is produced in the southern hemisphere from the nuclear explosions of which the authentic records are not available.

b) Iceface and ice core samples

Natural and artificial total beta activity in iceface samples collected near Indian Bay and two shallow ice cores (Tables II, III) have been measured. ^{137}Cs

Table V. ^{137}Cs and Total beta activity in fresh snow and shelf ice cores from Antarctica

Code No	Nature/ Depth (m)	Volume (L)	^{137}Cs activity (dph/L)	Total beta activity (dph/L)
A	Fresh Snow	4.34	11.40 ± 10.80	—
Shelf Ice Core (DG)				
G-1	0-0.6	1.65	BDL	83.67 ± 11.04
G-2	0.6-1.2	0.75	40.20 ± 58.80	174.60 ± 36.00
G-3	1.2-1.8	1.60	31.20 ± 25.20	126.00 ± 12.00
G-4	1.8-2.4	1.35	BDL	45.00 ± 10.20
Shelf Ice Core (Indian Bay)				
N-2	0-0.5	1.12	BDL	428.47 ± 34.59
N-3	0.5-1.0	0.95	BDL	110.80 ± 11.30
N-4	1.0-1.5	1.25	BDL	134.80 ± 12.70
N-8	3.0-3.5	1.38	BDL	125.60 ± 16.94
N-10	4.0-4.5	1.15	33.00 ± 13.8	116.47 ± 11.99
N-12	5.0-5.5	1.63	17.40 ± 11.40	95.30 ± 13.40
N-14	6.0-6.5	1.60	47.46 ± 10.02	233.63 ± 30.66
N-15	6.5-7.0	0.85	2.35 ± 18.72	257.40 ± 27.57
N-16	7.0-7.5	1.54	19.20 ± 10.20	108.57 ± 13.69
N-18	8.0-8.5	1.52	14.47 ± 6.58	238.80 ± 28.65

BDL = Below detection limit.

activity which is produced only due to past nuclear explosions has been measured in only selected 14 samples (Table V) in iceface and ice core samples for comparison with that of total beta measurements.

Both ^{137}Cs as well as total beta activity in 9m iceface samples (Table III and V) show a sharp peak at 6–6.5m depth (Fig. 2) corresponding to 1961-62 Russian nuclear test estimating an accumulation rate of $13.5\text{g/cm}^2 \cdot \text{yr}$ (0.3 m/yr)

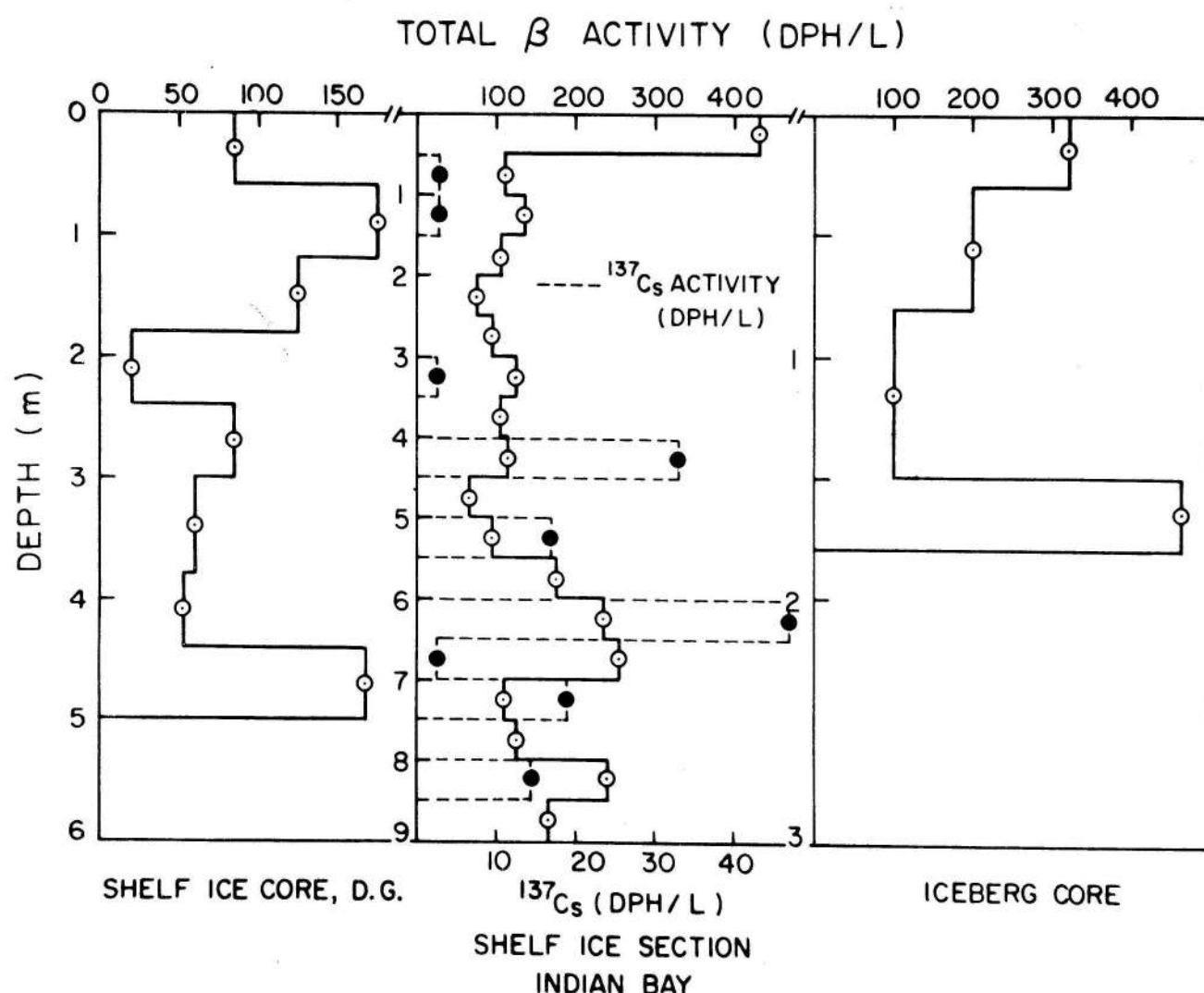


Fig. 2. Total beta activity (dph/L) of the shelf ice core, shelf ice section Indian Bay and iceberg has been plotted as a function of depth. The shelf ice core near D.G. shows no systematic variation with depth whereas the iceberg core shows a high value at 1.5–2m (see text). The total beta activity and ^{137}Cs profiles show a sharp peak activity at 6–6.5m depth indicating the presence of nuclear horizon of 1961-62 in that depth fraction. This indicates the past accumulation rate of 0.3m/yr at this location. For explanation of other peaks, see text.

assuming an average density of 0.45 g/cm^3 . ^{137}Cs shows another peak at 4–4.5m probably corresponding to 1969–70 French nuclear test. Moreover the total beta activity shows additional two peaks at 8–8.5m and 0–0.5m. The peak value at 8–8.5m corresponds to 1957–58 whereas the peak observed at 0–0.5m and similar high values observed in two very shallow sections of iceberg core at 0–0.5 and 1.5–1.8m is intriguing. To conclude, the data obtained on ^{137}Cs and total beta activities in samples collected from shallow depths of an iceface and ice core give a rough estimate of accumulation rate of ice as 0.3m/yr ($13.5 \text{ g/cm}^2 \text{ yr}$ of water equivalent assuming an average density of 0.45gm/cm^3) near Dakshin Gangotri Station which reasonably agrees with that observed by ^{210}Pb method and the knowledge of precipitation in that region.

Suggestion

It is desirable to drill long ice cores upto 50m in the shelf ice and polar ice to identify the nuclear horizons deposited during past four decades which may be used as time-markers to study past accumulation rates of ice, climatic and pollution records.

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